

$$\underline{\underline{\sigma}} = \underline{\underline{C}} : \underline{\underline{\epsilon}}$$

$\sigma_{ij} = \lambda e_{kk} \delta_{ij} + 2\mu e_{ij}$ - Material clásico lineal

$$\tau_{ij} = C_{ijkl} E_{kl}$$

$$\underline{\underline{\tau}} \cdot \underline{\underline{v}} = \underline{\underline{z}} \underline{\underline{v}}$$

$$\underline{\underline{\tau}} \cdot \underline{\underline{v}} - \underline{\underline{z}} \underline{\underline{v}} = 0$$

$$\underline{\underline{\tau}} \cdot \underline{\underline{v}} - \underline{\underline{z}} \underline{\underline{\tau}} \cdot \underline{\underline{v}} = 0$$

$$(\underline{\underline{\tau}} - \underline{\underline{z}} \underline{\underline{\tau}}) \underline{\underline{v}} = 0$$

$$(\underline{\underline{\tau}} - \underline{\underline{z}}^{(k)} \underline{\underline{I}}) \underline{\underline{v}}^{(k)} = 0$$

direcciones principales de tension

$$(\tau_{ij} - z^{(k)} \delta_{ij}) v_j^{(k)} = 0$$

$$\tau_{ij} = \lambda E_{kk} \delta_{ij} + 2\mu E_{ij}$$

$$(\lambda E_{kk} \delta_{ij} + 2\mu E_{ij} - z^{(k)} \delta_{ij}) v_j^{(k)} = 0$$

$$\left[E_{ij} - \underbrace{\left(\frac{z^{(k)} - \lambda E_{kk}}{2\mu} \right)}_{\epsilon^{(k)}} \right] v_j^{(k)} = 0$$

$$\epsilon^{(k)} = \frac{z^{(k)} - \lambda E_{kk}}{2\mu}$$

$$v_j^{(k)} = \eta_j^{(k)}$$

los direcciones ppales de deformación y de tensión son los mismos

$$(\underline{\underline{E}} - \underline{\underline{\epsilon}}^{(k)} \underline{\underline{I}}) \underline{\underline{\eta}}^{(k)} = 0$$

direcciones ppales de deformación

$$(E_{ij} - \epsilon^{(k)} \delta_{ij}) \eta_j^{(k)} = 0$$

